# LATE BALIRAM KASHYAP GOVERNMENT MEDICAL COLLEGE

DIMRAPAL, JAGDALPUR

ACADEMIC YEAR
2026 – 27

## DEPARTMENT OF PHYSIOLOGY

## PROJECT ON

# LIFE-SAVING MODULES


**SUBMITTED BY:**

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MBBS 1st Year

Batch: 2026

**SUBMITTED TO:**

Department of Physiology,

LT. B.R.K.M. GMC, Jagdalpur

# CERTIFICATE

This is to certify that **Harsh Kumar Saw**, student of MBBS Batch 2026, LT. B.R.K.M. Government Medical College, Jagdalpur, has successfully completed the project entitled **"LIFE-SAVING MODULES"** under the Department of Physiology, during the academic year 2026 – 2027.

This project has been completed under the guidance and supervision of the Department of Physiology as a part of the academic curriculum.

Date: _____

Place: Jagdalpur

**Signature of Faculty Guide**

Name: _____

Designation: _____

Department of Physiology

**Signature of Head of Department**

Name: _____

Department of Physiology

LT. B.R.K.M, GMC, Jagdalpur

# ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Lt. B.R.K.M. Government Medical College, Jagdalpur**, for providing me the opportunity to undertake this project on **"Life-Saving Modules"** under the Department of Physiology.

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I would also like to extend my sincere thanks to my respected **Professor** and **all the faculty members of the Department of Physiology** for their valuable guidance, encouragement and assistance, which helped me in completing this project successfully.

Lastly, I am grateful to everyone who directly or indirectly contributed to the completion of this project.

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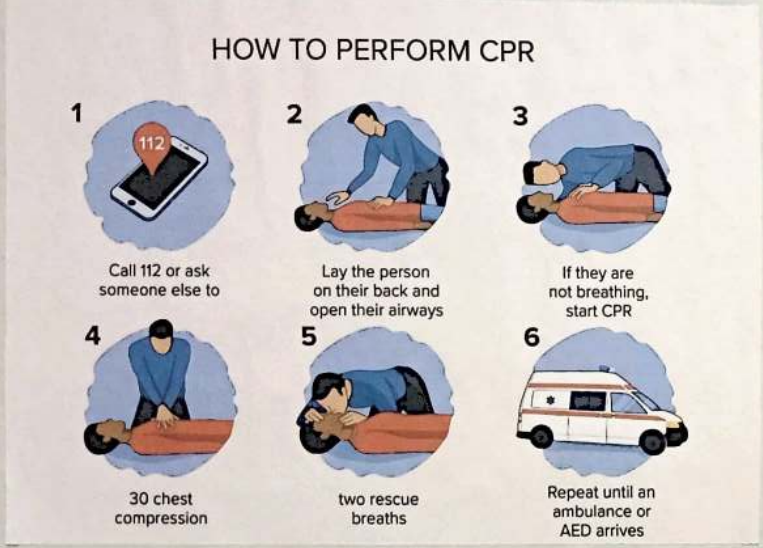
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# Cardiopulmonary Resuscitation (CPR)

## 1. Definition

⇒ Cardiopulmonary resuscitation (CPR) is an emergency procedure involving chest compressions, with rescue breaths when appropriate to maintain blood circulation and oxygen delivery to vital organs during cardiac arrest.

## 2. Objectives of CPR

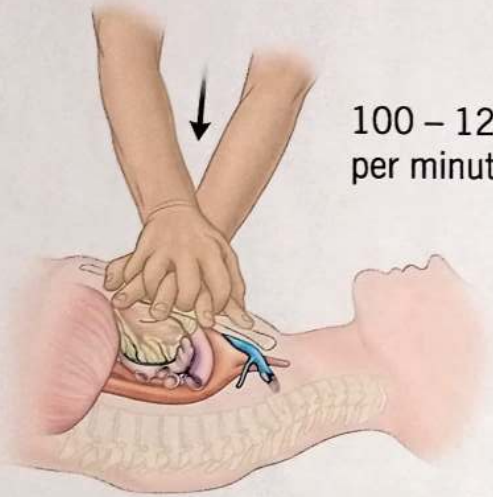
- To maintain blood flow to the brain and other vital organs
- To provide temporary circulation until advance medical care.
- To increase the chances of survival following cardiac arrest.
- To support early defibrillation when indicated.

## 3. Indications

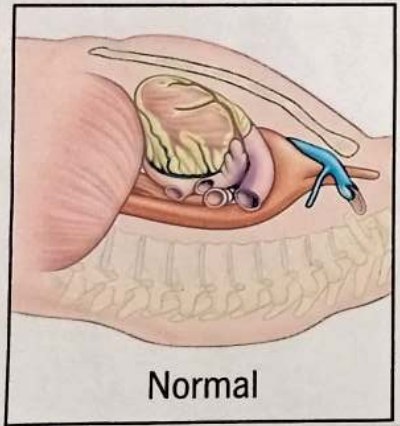
⇒ CPR is indicated when a person is unresponsive and is not breathing normally, including when only occasional gasping is present.

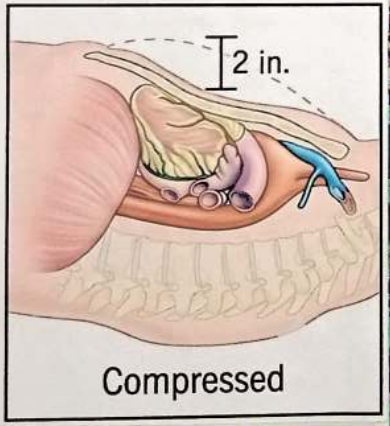
A lay rescuer should not delay CPR to check the pulse.

# Cardiopulmonary resuscitation (CPR)



100 – 120 compressions per minute






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4. Stages of CPR

1. Ensure safety: Check that the surrounding area is safe.

2. Check responsiveness: Tap the shoulder and ask loudly whether the person is all right

3. Call for help: Activate emergency medical services and ask someone to bring an automated external defibrillator (AED)

4. Assess breathing: Check for normal breathing for no more than 10 seconds

5. Position the patient: Place the person on their back on a firm, flat surface

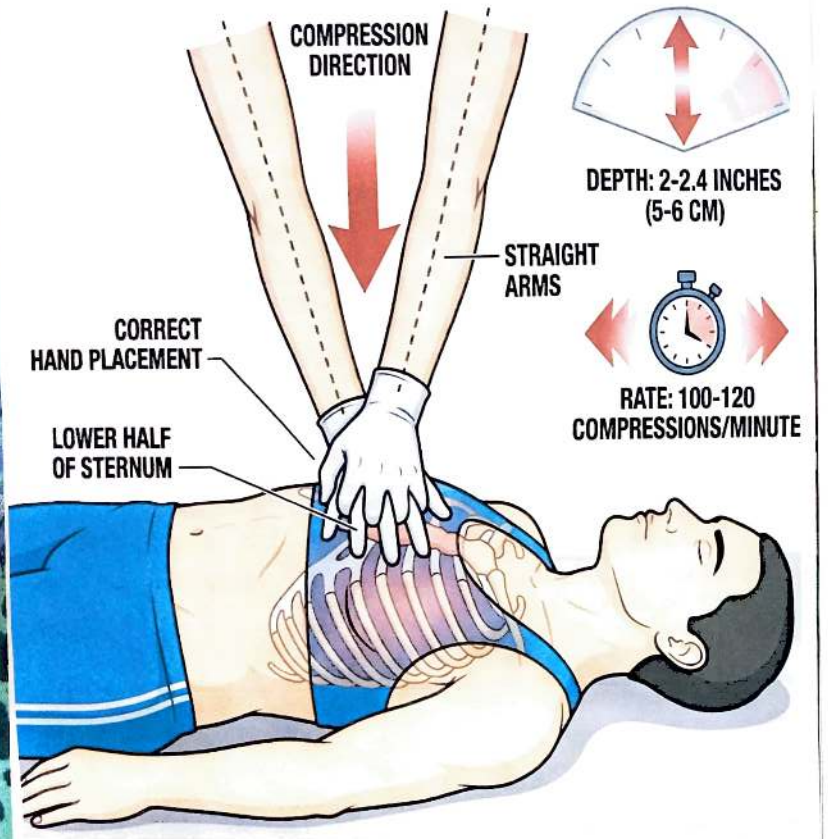
6. Perform chest compressions: Place the heel of one hand on the centre of the chest, on the lower half of the sternum, and the other hand over it.

7. Give rescue breaths: If trained and willing, provide 30 compressions followed by 2 breaths

8. Use AED: Apply it as soon as available and follow its instructions.

9. Continue resuscitation: Continue until the person shows

# STANDARD CPR TECHNIQUE GUIDE






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Signs of life and breathes normally, trained help takes over, the scene becomes unsafe, or you cannot continue.

5. Characteristics of high-quality CPR

- Compression rate : 100-120 per minute
- Compression depth : 5-6 cm
- Compression-to-ventilation ratio: 30:2 for a single rescuer
- Allow complete chest recoil after each compression.
- Minimize interruptions in chest compressions

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# Heimlich manoeuvre

## 1. Definition

⇒ The Heimlich manoeuvre, also known as abdominal thrusts, is a first-aid technique used to relieve severe airway obstruction caused by a foreign body in a conscious adult or child older than one year.

## 2. Causes of choking

⇒ Choking occurs when a foreign body partially or completely obstructs the airway. Common causes include:

- Food particles such as meat or nuts.
- Small objects accidentally entering the airway
- Poorly chewed food
- Eating too quickly.

## 3. Signs of severe choking

- Inability to speak or breathe.
- Ineffective or absent coughing
- Clutching the throat
- Development of bluish discoloration in severe cases






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- Loss of consciousness if obstruction persists

4. Stages of the Heimlich manoeuvre

1. Recognise choking: Ask the person if they are choking.

2. Encourage coughing: If the person can cough forcefully, encourage coughing and observe closely.

3. Call for help: If coughing is ineffective or the person cannot speak or breathe, activate emergency services.

4. Give back blows: Lean the person forward and deliver 5 firm back blows between the shoulder blades.

5. Give abdominal thrusts: If the obstruction persists, stand behind the person, place a fist just above the navel and below the breastbone, grasp it with the other hand, and give 5 quick inward-and-upward thrusts.

6. Repeat: Alternate 5 back blows and 5 abdominal thrusts until the obstruction is relieved or the person becomes unresponsive.

7. If unconscious: Lower the person safely, call emergency services and begin CPR. Remove an object from the mouth only if it is clearly visible.

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Important precautions

- Do not use abdominal thrusts on infants under one year; use infant-specific back blows and chest thrusts.
- For pregnant people or when the abdomen cannot be encircled, use chest thrusts according to current first-aid guidance.
- Medical assessment may be needed after abdominal thrusts, particularly if pain or other symptoms persists.

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